

Resit Assignment 4

TU Delft, ME, DCSC, Spring 2025

- The assignments are individual. You may consult and discuss with your colleagues, but the answers you provide, the code you use, and the report you hand in must be independent.
- Provide detailed answers, describing the steps you followed. Only end results of calculations are not sufficient.
- Provide a clear report as a PDF file, typed, preferably using L^AT_EX.
- Submit the report digitally via Brightspace before the deadline: 9:00, September 1, 2025.
- You may use Matlab/Python to solve the assignment.
- Include your code, for reproducibility of your results, in your Brightspace submission on a zip file, or through a link on your report.
- The code will only be used to verify reproducibility in case of doubts. The grading will be performed based on the documentation and results described in the report.

(4+1)p Problem 1

Similarly to what has been seen in the lectures, we would like to solve a multi-aircraft coordination problem, where each aircraft i can be described by a linear time-invariant dynamical system

$$x_i(t+1) = Ax_i(t) + Bu_i(t), \quad x_i(0) = x_{i,0}, \quad i = 1, \dots, 4, \quad t = 0, \dots, T_{\text{final}} - 1.$$

The objective is to find a decomposition-based algorithm for coordinating towards a common target state at T_{final} , i.e., to ensure that

$$x_i(T_{\text{final}}) = x_f, \quad \forall i$$

while satisfying an element-wise limitation on the control inputs

$$|u_i(t)| \leq \frac{u_{\text{max}}}{T_{\text{final}}}, \quad \forall i, t.$$

The objective function to be minimized is the following quadratic function (which includes also $x_i(T_{\text{final}})$)

$$\sum_i \sum_t x_i(t)^T x_i(t) + u_i(t)^T u_i(t).$$

The optimization variables are the (private) controls u_i of the individual aircraft, as well as the (public) common terminal state x_f . A Matlab file is attached (**aircraft.m**) with the matrices A_i and B_i along with the initial states, control limit, and horizon length for a four-aircraft example.

- 1p (a) Derive a solution based on dual decomposition using the projected subgradient method and demonstrate it with the use of Matlab or Python (use of cvx toolbox is allowed). Show a plot that demonstrates the convergence of the optimization process (in terms of a logarithmic error sequence). Show a plot of the resulting aircraft state trajectories after convergence, and compare with the centralized optimal solution.
- 1p (b) Investigate the effect of step size (for constant step) and step size update sequence (for variable step) on the convergence of the standard subgradient method. Show the results using a logarithmic error sequence plot.
- 1p (c) Implement an accelerated version (e.g., Nesterov method) of the subgradient updates. Show the results using a logarithmic error sequence plot.
- 1p (d) Consider the standard incremental subgradient method, and the following consensus matrix between the agents:

$$W = \begin{pmatrix} 0.75 & 0.25 & 0 & 0 \\ 0.25 & 0.5 & 0.25 & 0 \\ 0 & 0.25 & 0.5 & 0.25 \\ 0 & 0 & 0.25 & 0.75 \end{pmatrix}$$

Implement a combined consensus/incremental subgradient approach and investigate the effect of the number of consensus steps in each subgradient iteration on the convergence rate.

- 1p (e) For extra points (and a good learning experience), consider replacing the linear input constraint with the following quadratic limitation on the total control energy

$$\sum_t u_i(t)^T u_i(t) \leq \frac{2u_{\max}^2}{T_{\text{final}} + 2}, \quad \forall i.$$

You may need to have access to solvers that are able to solve the resulting quadratically constrained quadratic programs (QCQP), such as Yalmip using the Gurobi solver. Show a plot of the resulting aircraft state trajectories and control inputs after convergence, and indicate the effect of the quadratic input constraints.

3p Problem 2

- 2p (a) Consider the multi-aircraft coordination formulation in Problem 1 and implement a consensus ADMM approach (see ADMM lecture slides) to solve this problem. Show a plot that demonstrates the convergence of the optimization process (in terms of a logarithmic error sequence). Show a plot of the resulting aircraft state trajectories after convergence, and compare with the centralized optimal solution.
- 1p (b) Investigate the effect of the ρ parameter on the convergence rate. Show the results using a logarithmic error sequence plot.